Ellipses

ANSWER KEY

Standard form

1 For the ellipse $\frac{x^2}{25}+\frac{y^2}{9}=1$, find:

a the vertices $(\pm 5, 0)$

b the foci $(\pm 4, 0)$

c the endpoints of the minor axis $(0, \pm 3)$

d the eccentricity $e=\frac{4}{5}$

2 Write the standard-form equation of the ellipse with foci $(0, \pm 3)$ and vertices $(0, \pm 5)$.

Major axis vertical with $a=5$, $c=3$, so $b^2=25-9=16$: $\frac{x^2}{16}+\frac{y^2}{25}=1$.

3 Consider the ellipse $9x^2+4y^2=36$.

a Write the equation in standard form. $\frac{x^2}{4}+\frac{y^2}{9}=1$.

b State the lengths of the major and minor axes. Major axis $2a=6$ (vertical); minor axis $2b=4$.

c Find the coordinates of the foci. $c^2=9-4=5$, so foci $(0, \pm\sqrt{5})$.

Applications

4 A whispering gallery has an elliptical ceiling 40 m long and 24 m high at the center (so $a=20$ and $b=12$, with the major axis horizontal). Two people stand at the foci. How far apart are they, to one decimal place?

$c=\sqrt{400-144}=\sqrt{256}=16$, so the foci are $2c=32.0$ m apart.

5 An ellipse centered at the origin has a horizontal major axis of length 12 and passes through the point $(3, 2\sqrt{3})$. Find its equation.

$a=6$: $\frac{9}{36}+\frac{12}{b^2}=1$ gives $\frac{12}{b^2}=\frac{3}{4}$, so $b^2=16$: $\frac{x^2}{36}+\frac{y^2}{16}=1$.